Matthew Kim

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Education

University of Virginia Charlottesville, VA
B.A. in Applied Statistics, Minor in Computer Science | GPA: 3.9/4.0 *Aug. 2023 – May 2027*

- **Relevant Coursework:** Statistical Machine Learning, Data Visual and Management, Data Analysis with Python, Design & Analysis of Sample Surveys, Intro Mathematical Statistics, From Data to Knowledge, Intro to Regression Analysis, Software Dev Essentials, Computer Systems and Org 1, Discrete Math and Theory 1, Data Structures and Algo 1

Experience

Research Assistant 2024 – Present
UVA Historic Fashion Archive *Charlottesville, VA*

- Architected a Next.js 16 / React 19 / TypeScript frontend for UVA's historic clothing collection on a CollectiveAccess CMS (custom API client, 5-minute cache, static JSON fallback), with multi-attribute filtering, Boolean search, JSON/CSV/PDF export, and a Three.js 3D garment viewer
- Rebuilt the related-item recommender as a learned re-ranker over FashionCLIP image embeddings and extracted attributes, raising nDCG@5 from 0.70 to 0.93 (95% CI +0.15 to +0.31) on a 50-query blind-judged benchmark under nested cross-validation
- Deployed collection search fusing BM25 with FashionCLIP text-to-image retrieval as a CPU-only FastAPI service (nDCG@5 0.80 vs. 0.68 for keyword search), and prototyped a local Qwen3-8B query parser with citation-checked RAG answers (nDCG@5 0.88); gated by GitHub Actions and tracked in MLflow
- Extracted material, color, and technique attributes from 1,400+ catalog records (F1 0.95–0.99) and trained a linear probe on weak labels to tag undescribed garments from photos (color F1 0.58 vs. 0.39 zero-shot)
- Piloted a 3D shape-descriptor pipeline (curvature, symmetry, drape via `trimesh`) on photogrammetry scans, applying PCA and clustering evaluated against catalog metadata (ARI, NMI)

Teaching Assistant 2026 – Present
University of Virginia — CS1112 *Charlottesville, VA*

- Supported 300+ introductory Python students through weekly office hours, lab sections, and one-on-one debugging, and graded assignments and exams with structured feedback

Projects

Mortgage Default Risk Model | *Python, pandas, scikit-learn, LightGBM* 2026

- Modeled 24-month default for 600K+ Fannie Mae loans (106 GB of performance data); logistic regression, random forest, and gradient boosting tied at ROC AUC 0.83 on an out-of-time test set
- Traced a 5–8× Florida/Texas default spike to 2017 hurricane forbearance and redefined the label; at 100:1 costs, the calibrated model flags 7% of loans and catches 39% of defaults

Hana-Bi: E-Commerce Platform | *Next.js, TypeScript, Stripe, Resend, Zustand, Framer Motion* 2024 – Present

- Engineered a full-stack e-commerce platform with Stripe as the product source of truth: an autopaging catalog pipeline, persisted cart state, Checkout Sessions, and webhook-driven order and waitlist emails via Resend

Tech Industry Layoff Analysis | *R, tidyverse, ggplot2, R Markdown* 2026

- Analyzed 2,700+ tech layoffs with t-tests, ANOVA, and ggplot2 across company types and industries, finding that private companies cut twice the share of their workforce that public ones did (32.4% vs. 16.7%)

Insurance Client Data Management System | *Python, Flask, SQLite, Google Sheets API, Render* 2025 – 2026

- Built a Flask dashboard that syncs the agency's Google Sheet into a read-only SQLite copy for search, sorting, and CSV export, replacing a binder-based workflow covering ∼2,000 records; rebuilt in 2026 with a 30/60/90-day renewals view, an audit log, and a pytest suite in GitHub Actions

NBA Salary Prediction Model | *R, multiple linear regression, cross-validation* 2025

- Built a multiple linear regression model of NBA player salaries (adj. $R^2 \approx 0.62$, matched by five-fold cross-validation), with full EDA, variable selection, assumption diagnostics, and Cook's distance leverage checks; Points, Minutes, Stocks, and Age were the key predictors

Technical Skills

Languages: Python, R, Java, C, JavaScript, TypeScript, SQL
Machine Learning & AI: PyTorch, scikit-learn, LightGBM, OpenCLIP (FashionCLIP), Ollama, MLflow, RAG, Generative AI, retrieval & ranking (BM25, nDCG), MLOps/LLMOps
Frameworks & Tools: React, Next.js, FastAPI, Flask, pandas, NumPy, SciPy, tidyverse, ggplot2, plotly, R Shiny, rvest (web scraping), Git/GitHub, GitHub Actions (CI/CD), Stripe API, REST APIs, Jupyter
Statistical Methods: Hypothesis testing, ANOVA, linear & logistic regression, ridge/lasso regularization, KNN, LDA, decision trees, random forests, gradient boosting, splines, bootstrap, Monte Carlo simulation, cross-validation, ROC/AUC, precision/recall, EDA, Model Evaluation
Survey Sampling: Stratified & cluster sampling, sample-size determination, ratio & regression estimation, capture-recapture